

Fatma Amr Abdelrahman

AI Software Engineer

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[Portfolio](#)

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Education

Bachelor's Degree – Faculty of Computers and Artificial Intelligence, Capital University 2022 - 2026
GCI World (Global Consumer Intelligence), Data Science and AI, The University of Tokyo April 2026 - July 2026

Professional Experience

Duckurity – AI Engineer Intern

August 2026 – Present

- Contributed to AI and cybersecurity projects, focusing on agentic workflows and AI-powered backend systems.
- Built multi-step reasoning and dynamic routing loops using LangGraph, enabling the assistant to call external tools and retrieve context reliably.
- Developed asynchronous FastAPI backend endpoints to integrate the assistant into the website with structured, low-latency responses.

Technical Skills

- AI & ML:** Machine Learning (Scikit-learn), Deep Learning (PyTorch, TensorFlow), NLP, LLMs, RAG, Prompt Engineering, Model Evaluation, LangChain, LangGraph, Vector Databases (Qdrant, Pinecone, Chroma)
- Programming:** Python, C# (.NET), OOP, Data Structures & Algorithms, HTML/CSS, Frontend Integration
- Backend & Databases:** FastAPI, REST APIs, SQL, Relational Database Design
- Tools & Infrastructure:** Git, GitHub, Docker, Apache Kafka, RabbitMQ, Postman, PyTest, API Testing
- Engineering:** Technical Documentation, API Integration, Agile Development

Trainings

- Samsung Innovation Campus – AI Apprenticeship** July 2026 – October 2026
- Genius Technology Center (GTC) - Machine learning Interns Ambassador** August 2025 – September 2025
- ALX Africa - Data Science Trainee** January 2025 – November 2025
- Elevvo Pathway - Machine Learning Intern (Outstanding Contributor)** January 2025 – September 2025
- National telecommunication institute (NTI) - Computer Vision Trainee** July 2025 – August 2025
- National telecommunication institute (NTI) - AI engineering Trainee** August 2024 – September 2024
- Digital Egypt pioneers Initiative (DEPI) - Backend (.NET) web developer** May 2024 – October 2024

Projects

Asset Sentinel (Industrial Predictive Maintenance Platform) | *FastAPI, Llama 3, RabbitMQ, RAG* - [Repo](#)

- Engineered** a multi-tenant Predictive Maintenance (PdM) platform utilizing a **3-Layer Stacked Deep LSTM** network to execute real-time Remaining Useful Life (RUL) forecasting.
- Implemented** a **Piecewise Linear RUL Target Strategy** ($\text{clip}(\text{upper}=125)$) over stateful sliding context windows ($W=64$), mitigating early-stage gradient noise and achieving an **RMSE of 7.551** and an **R2 score of 0.9654** on NASA's CMAPSS dataset.
- Architected** a high-throughput, decoupled ingestion pipeline using **RabbitMQ** and **FastAPI asynchronous workers** to stream and log multi-dimensional telemetry arrays into **MongoDB**.
- Built** an **Interactive Executive Dashboard** integrated with a natural language **Text-to-SQL execution engine**, empowering operators to query live database streams using fluid, plain-text queries.

ELARA (Clinical-Grade RAG System) | *FastAPI, Qdrant, Docling, MongoDB* - [Demo](#) / [Repo](#)

- Engineered an end-to-end clinical RAG pipeline to process complex, medical literature and unstructured documents using **Docling** for layout-aware PDF extraction.
- Implemented dense semantic retrieval over **Qdrant** vector database leveraging **BAAI/bge-m3** embeddings, optimized with chunking strategies to minimize factual hallucinations.
- Built deterministic guardrails and safety verification gates ensuring domain-specific response alignment before surfacing outputs to the end user.
- Delivered high-performance asynchronous query endpoints using **FastAPI** with structured metadata filtering and MongoDB document storage.

SwiftApply (CV Generation Automator) | *FastAPI, Prompt Engineering, Llama 3, LangChain* - [Demo](#) / [Repo](#)

- Engineered a **modular backend** using **FastAPI** to automate the end-to-end process of generating ATS-compliant CVs tailored to specific Job Descriptions.
- Developed complex **form-handling logic** and **data validation workflows** to manage multi-source user inputs (Candidate Profiling & JD Parsing).
- Integrated **Llama 3** via advanced **Prompt Engineering** to automate keyword extraction and role-specific content rewriting.
- Delivered an end-to-end pipeline that reduces manual CV editing time and helps job seekers generate customized applications in one click.

NDR & Real-Time SOC Triage System | *Python, Kafka, XGBoost, Scikit-Learn, FastAPI* - [Demo](#) / [Repo](#)

- A real-time Network Detection and Response (NDR) pipeline designed as an automated SOC triage tool to mitigate behavioral blindness and alert fatigue in enterprise environments.
- Engineered a decoupled, streaming ingestion engine using **Apache Kafka** to consume and parse network telemetry and packet flows simulated on the UNSW-NB15 dataset.
- Implemented a **two-stage machine learning inference pipeline** (Random Forest & XGBoost) to execute real-time binary anomaly filtering followed by granular, multi-class threat classification.
- Built automated risk-scoring and triage logic to rank incidents by urgency, generating actionable forensic metadata and threat severity indicators for Tier-1 SOC analysts.

Volunteering & Leadership

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| • Google Developer Group (GDG AI-Azhar) - Data Science Lead | November 2025 – present |
| • ALX Voltra – Public Relations (PR) Specialist in Events Organization | April 2025 – present |

Courses & Certificates

- **LLMOps Concepts** – *DataCamp*
- **Introduction to Vector Databases with Pinecone** – *365 Data Science*
- **Getting Started with Deep Learning** – *NVIDIA*
- **Introducing Generative AI with AWS** – *Udacity / AWS*
- **Introduction to Apache Kafka** – *DataCamp*